

EE446_Lab_8

For the software part, I tried changing the MSE to accuracy as it would help to develop the model based on accuracy metric than other methods. So, by changing that it was able to compile the model and later give us the right plots for it. For plotting the training and validation accuracy, I also changed the metric to plot based on accuracy then what it was written in past which wasn't giving us the right answer.



Screenshot of serial monitor:

I tried very hard to show the sup classification to be close to .99 but for some reason the board IMU wasn't giving it but moving the board left to right gave us the HI stage around 99% of the time.

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Arduino Nano 33 BLE

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lab9-classifier-dual-board.ino model.h

```
14 Board selection:
15 - Set this to 1 for Arduino Nano 33 BLE Sense Rev2 / Nano 33 BLE Rev2.
16 - Set this to 0 for the original Arduino Nano 33 BLE Sense / Nano 33 BLE boards
17   included with older TinyML kits.
18
19 The rest of the sketch can stay the same because both IMU libraries expose
20 the same IMU object and the same basic accelerometer/gyroscope API.
21 */
22 #define USE_NANO_33_BLE_REV2_IMU 0
23
24 #if USE_NANO_33_BLE_REV2_IMU
25 #include <Arduino_PMT270_PMM150.h>
```

Output Serial Monitor X

Message (Enter to send message to 'Arduino Nano 33 BLE' on 'COM10') New Line 115200 baud

```
21:08:53.189 ->
21:08:54.292 -> hi: 0.998472
21:08:54.292 -> sup: 0.001528
21:08:54.292 ->
21:08:55.435 -> hi: 0.998998
21:08:55.472 -> sup: 0.001002
21:08:55.472 ->
21:08:56.491 -> hi: 0.981106
21:08:56.491 -> sup: 0.018894
21:08:56.491 ->
21:08:59.988 -> hi: 0.970659
21:08:59.988 -> sup: 0.029341
21:08:59.988 ->
21:09:04.928 -> hi: 0.977974
21:09:04.928 -> sup: 0.022026
21:09:04.973 ->
21:09:06.899 -> hi: 0.448655
21:09:06.899 -> sup: 0.551345
21:09:06.899 ->
21:09:07.900 -> hi: 0.687685
21:09:07.900 -> sup: 0.312315
21:09:07.900 ->
```

Ln 35, Col 1 Arduino Nano 33 BLE on COM10

9:09 PM 8/15/2026